

STACKFORGE

Forging India's Elite Developers

8-Month Full Stack, DSA & AI Mastery Program

Curriculum Guide · July 2026 Batch

8

Months

5+

Live Projects

141

Teaching Days

100%

Job Focused

3

Mock Interviews

WHAT YOU'LL MASTER

● Full Stack (MERN)

● DSA & Algorithms

● System Design

● Generative AI

● Cloud & DevOps

● Interview Prep

PROGRAM TIMELINE

M1

Foundations

M2

React

M3

Backend

M4

Full Stack

M5

GenAI

M6

Sys Design

M7

Capstone

M8

Interview

WHAT YOU'LL DELIVER EACH MONTH

M1 Deploy your first full stack web app

M2 Ship a polished React SPA

M3 Deliver a production-grade REST API

M4 Complete authenticated MERN application

M5 Integrate working AI features

M6 Deploy containerised app to cloud

M7 Production-grade AI capstone live

M8 Demo Day · Land your dream offer

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Month 1

Foundations & Web Essentials

Weeks 1–4

PHASE 1 OF 8

FULL STACK DEVELOPMENT

- HTML5 & CSS3 — Semantic, Flexbox & Grid
- CSS Animations, Variables & Dark Mode
- JavaScript ES6+ — Functions & Closures
- DOM Manipulation, Events & Async/Await
- Git, GitHub & Terminal Workflow

DSA & PROBLEM SOLVING

- Complexity Analysis — Big-O Notation
- Arrays & Strings — Core Patterns
- Two Pointer & Sliding Window
- Hashing & Frequency Map Patterns
- Prefix Sum & Kadane's Algorithm

MONTH OUTCOME

Build & deploy your first full stack web app

2

Month 2

React & Frontend Engineering

Weeks 5–8

PHASE 2 OF 8

FULL STACK DEVELOPMENT

- React Core — JSX, Components & Props
- Hooks — useState, useEffect & Custom Hooks
- React Router, Forms & Validation
- State — Context, useReducer & Redux Toolkit
- React Query, Performance & Code Splitting

DSA & PROBLEM SOLVING

- Linked Lists — Singly, Doubly & Circular
- Stacks & Queues — Monotonic Stack
- Binary Trees — DFS/BFS Traversals
- BSTs & Heaps (Priority Queue)
- Graphs — BFS, DFS & Topological Sort

MONTH OUTCOME

Ship a polished React SPA with real-world patterns

3

Month 3

Backend & Database Engineering

Weeks 9–12

PHASE 3 OF 8

● FULL STACK DEVELOPMENT

- Node.js — Architecture, Streams & Events
- Express.js — Routing, Middleware & JWT
- API Security — Helmet, Rate Limit & OWASP
- MongoDB — CRUD, Aggregation & Indexing
- Mongoose — Schemas, Hooks & Transactions

● DSA & PROBLEM SOLVING

- Graphs — Dijkstra & Union-Find
- Dynamic Programming — 1D Patterns
- Dynamic Programming — 2D & String DP
- Backtracking — Subsets, Perms & N-Queens
- Greedy Algorithms & Interval Scheduling

MONTH OUTCOME

Deliver a production-grade REST API with database

4

Month 4

Full Stack Integration & Auth

Weeks 13–16

PHASE 4 OF 8

● FULL STACK DEVELOPMENT

- JWT Refresh Tokens & OAuth 2.0
- Real-time Features with Socket.io & SSE
- File Uploads, Cloudinary Storage
- SQL — Joins, CTEs & Window Functions
- API Testing — Jest & Supertest

● DSA & PROBLEM SOLVING

- Bit Manipulation — XOR & Bitmasking
- Sorting — Merge, Quick & Counting Sort
- Tries, Segment Trees & Fenwick Tree
- Hard Problems Sprint — FAANG Patterns
- Mock Test 1 — Full DSA Assessment

MONTH OUTCOME

Complete authenticated full stack MERN application

5

Month 5

Generative AI & CS Fundamentals

Weeks 17–20

PHASE 5 OF 8

● FULL STACK DEVELOPMENT

- OpenAI API & Prompt Engineering
- LangChain Fundamentals & RAG Pipelines
- Vector Databases & Semantic Search
- Computer Networks — TCP/IP, HTTP & DNS
- Operating Systems — Threads & Memory

● DSA & PROBLEM SOLVING

- Sorting & Arrays Advanced Patterns
- Graph Sprint — SCCs & Bellman-Ford
- DP Sprint — Knapsack, LCS & Edit Distance
- Company Sprints — Amazon & Microsoft
- Mock Test 2 — Mid-Course Assessment

MONTH OUTCOME

Integrate a working AI feature into a live product

6

Month 6

System Design, LLD & DevOps

Weeks 21–24

PHASE 6 OF 8

● FULL STACK DEVELOPMENT

- HLD — CAP Theorem & Scalability
- Caching (Redis), Sharding & Message Queues
- LLD — SOLID Principles & Design Patterns
- Case Studies — WhatsApp, YouTube & Uber
- Docker, CI/CD with GitHub Actions & AWS

● DSA & PROBLEM SOLVING

- Company Sprints — Google & Startups
- Final Revision — Trees, Graphs & DP
- Advanced Python — Async & Functional
- LRU/LFU Cache & Rate Limiter Coding
- Mock Test 3 — Final Full Assessment

MONTH OUTCOME

Deploy a containerised app to cloud with CI/CD

7

Month 7

Capstone Build & Advanced Topics

Weeks 25–28

PHASE 7 OF 8

FULL STACK DEVELOPMENT

- Capstone 1 — Backend, Auth & AI Routes
- Capstone 2 — React Frontend & AI Streaming
- Capstone 3 — Docker, CI/CD & Cloud Deploy
- Capstone 4 — Performance & Security
- Advanced — TypeScript & Microservices

DSA & PROBLEM SOLVING

- Interview Prep — JS & React Technical
- Interview Prep — Backend & Node.js
- CS Fundamentals — OS, CN & DBMS Revision
- Hard Problems Final Sprint
- LeetCode Mixed Hard & FAANG Sims

MONTH OUTCOME

Production-grade AI capstone deployed end-to-end

8

Month 8

Interview Preparation & Career Launch

Weeks 29–32

FINAL PHASE · CAREER LAUNCH

- ATS Resume, LinkedIn & GitHub Portfolio
- Company Playbooks — Google, Amazon & Meta
- Full Mock Interviews — Rounds 1, 2 & 3
- System Design Mock Rounds — Advanced
- Behavioural Rounds — STAR Method Mastery

- Complete Pattern Revision Sprint
- FAANG Hard Sets & Timed Simulations
- Peer Teaching & Knowledge Solidification
- Job Search, Cold Outreach & Negotiation
- Final Mock Test — Complete Simulation

MONTH OUTCOME

Present capstone at Demo Day · Land your offer